

## CHAPTER 1 RESULTS

### 1.1 Cohort Characteristics

The Cancer Genome Atlas (TCGA)-Lung squamous cell carcinoma (LUSC) datasets for five modalities were retrieved (described in Section ??).

Gene expression and Copy number variation (CNV) retained the full labelled 178-sample cohort, whereas Deoxyribonucleic acid (DNA) methylation and micro RNA (miRNA) expression had smaller assay overlap (Table 1.1). Across modalities, the classical subtype was the most frequent class and primitive the least frequent, motivating the use of macro-averaged classification metrics in subsequent analyses.

Table 1.1: TCGA-LUSC omics modalities used in this study. Variation in sample size reflects assay availability and label overlap.

| Modality         | Configuration                     | Classical | Basal | Secretory | Primitive | Total |
|------------------|-----------------------------------|-----------|-------|-----------|-----------|-------|
| Gene expression  | HiSeqV2, RNASeqV2 RSEM-normalised | 65        | 43    | 43        | 27        | 178   |
| DNA methylation  | HumanMethylation450k beta values  | 29        | 14    | 22        | 11        | 76    |
| miRNA expression | miRNA sequencing, log2(RPM + 1)   | 44        | 29    | 20        | 14        | 107   |
| RPPA             | Normalised protein abundance      | 38        | 28    | 28        | 18        | 112   |
| CNV              | GISTIC2 all-thresholded calls     | 65        | 43    | 43        | 27        | 178   |

### 1.2 Preprocessing Summary

Table 1.2: Preprocessing and feature-retention summary for each omics modality. Raw feature counts correspond to the input assay feature spaces. The MAD threshold is the score of the last retained feature in the processed matrix; for RPPA, where all 238 antibodies were retained, this value represents the minimum observed MAD among retained antibodies.

| Modality         | Raw features | Retained features | MAD threshold |
|------------------|--------------|-------------------|---------------|
| Gene expression  | 60,660       | 2,000             | 1.0808        |
| DNA methylation  | 485,577      | 2,000             | 0.2208        |
| miRNA expression | 1,881        | 200               | 0.5954        |
| RPPA             | 238          | 238               | 0.0692        |
| CNV              | 24,776       | 2,000             | 1.0000        |

- 1.3 Characterization of LUSC Subtypes**
  - 1.3.1 Clinical characteristics**
  - 1.3.2 Immune Microenvironment**
  - 1.3.3 Somatic Mutation Landscape**
- 1.4 Dataset Separability**
- 1.5 Performance Across Omics Modalities**
- 1.6 SHAP Feature Importance and Stability**
- 1.7 Cross-Model Feature Overlap**
- 1.8 Cross-Modality Feature Overlap**
- 1.9 Pathway Enrichment**